

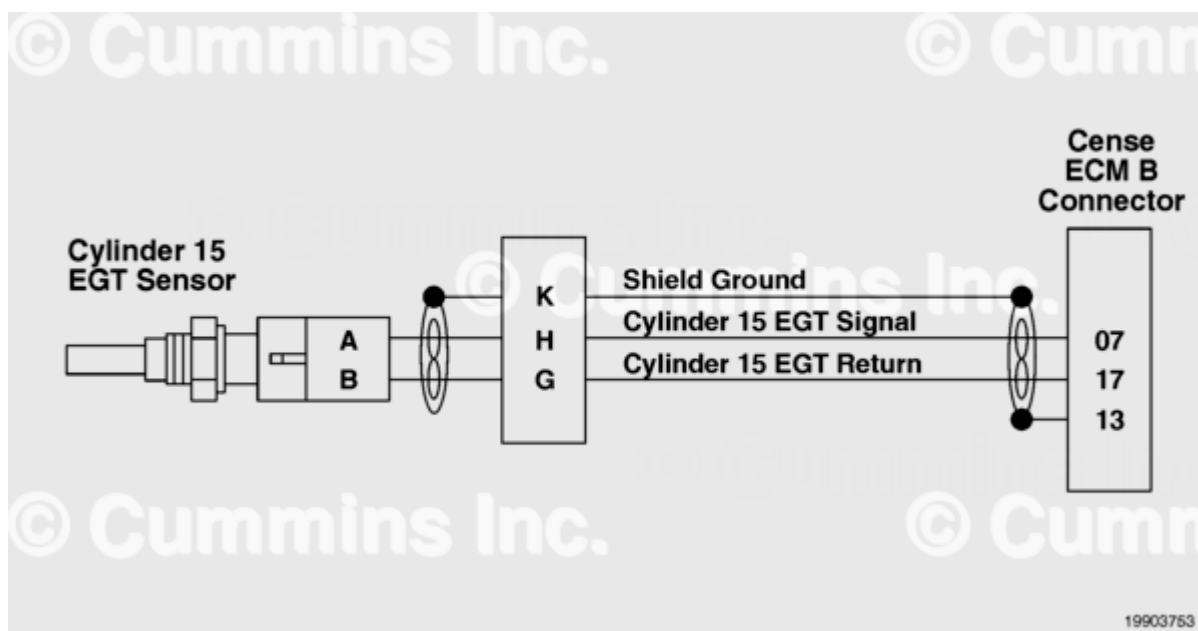
Fault Code: 678

Exhaust Gas Temperature Sensor Circuit Cylinder 15 (A8) - Voltage Below Normal or Shorted to Low Source

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Overview

Codes	Reason	Effect
Fault Code: 678 PID(P): SPN: 1151 FMI: 4/4 Lamp: Maintenance SRT: 00-391	Exhaust Gas Temperature Sensor Circuit Cylinder 15 (A8) - Voltage Below Normal or Shorted to Low Source. Low voltage detected on cylinder 15 exhaust temperature sensor SIGNAL pin 07 of the main harness B ECM connector.	Fault Codes 628, 648, and 668 are disabled.



Exhaust Gas Temperature Sensor Circuit Cylinder 15

Circuit Description

The exhaust gas temperature sensor circuit monitors exhaust gas temperature for cylinder 15 and passes information to the CENSE™ engine control module (ECM) through the engine harness. The ECM monitors the temperature and compares it to the exhaust gas temperatures of

other cylinders.

Component Location

The exhaust gas temperature sensor for cylinder 15 is located at the cylinder head-to-exhaust manifold interface.

Shoptalk

There are multiple CENSE™ ECMs for the engine models included in this manual. The ECM model displays when INSITE™ electronic service tool is connected. When troubleshooting a fault code, use the ECM model displayed in INSITE™ electronic service tool to determine which cylinder is affected. For engines with the present CM2330 ECM, the cylinder numbering sequence is described in the General Engine procedure of Section V in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 018-015 in Section V.

(/qs3/pubsys2/xml/en/procedures/56/56-018-015-tr.html)

The exhaust temperature sensor circuit is comprised of a reference thermocouple in the ECM and the exhaust temperature sensor thermocouple. The reference thermocouple in the ECM is at a known temperature and the exhaust temperature sensor thermocouple is at the exhaust port temperature. The voltage output of the circuit is directly related to the difference between the two thermocouple temperatures. With one temperature known, the ECM can calculate the exhaust temperature sensor thermocouple temperature.

Refer to Troubleshooting Fault Code t05-678 (/qs3/pubsys2/xml/en/procedures/81/81-t05-678.html)